

Engineering Mathematics III

ENGE702

Lecture 17: Laplace Transforms

Laplace Transforms

Now we will look at Laplace transforms of **derivatives**, and see how we might use them to solve differential equations.

Given a function $f(t)$, the following are the Laplace transforms of its derivatives:

$$\mathcal{L}(f'(t)) = s\mathcal{L}(f(t)) - f(0)$$

$$\mathcal{L}(f''(t)) = s^2\mathcal{L}(f(t)) - sf(0) - f'(0)$$

This means that initial conditions are built into Laplace transforms of derivatives of functions. This is one of the advantages of solving initial value problems this way: it is done directly.

Laplace Transforms

Let's see how we can use this to solve an initial value problem.

Example: Solve the following initial value problem:

$$y'' - y = t, \quad y(0) = 1, \quad y'(0) = 1$$

Laplace Transforms

Exercise: Use a Laplace transform to solve the initial value problem:

$$y' + 2y = 5 \cos(t), \quad y(0) = 0.$$

Laplace Transforms

Example: In electronics, in an RLC circuit, we can model the charge on the capacitor Q by the equation

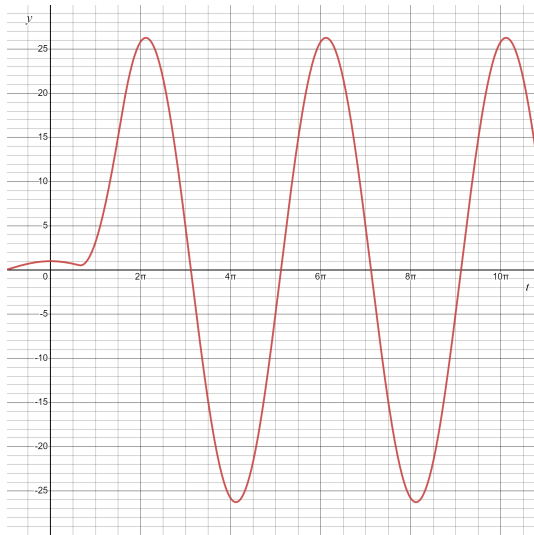
$$LQ'' + RQ' + \frac{1}{C}Q = E(t).$$

Suppose we have a circuit with $L = 1$, $R = 0$, $C = 4$, and $E(t)$ given by:

$$E(t) = \begin{cases} 5 & \text{if } 2 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

Suppose also the charge of the capacitor is 1 at time 0, and the rate of change in the charge is 0 at time 0. Find a function that represents the charge $Q(t)$.

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One final type of input function we will look at is called the Dirac delta function.

This is not a function in the technical sense, but it can be thought of as being 0 everywhere, except at one point where it is essentially infinite.

It will be useful for us in that it will allow us to model an instantaneous excitement of a system.

The **Dirac delta** function that acts at time $t = a$ is denoted by $\delta(t - a)$.

We can find the Laplace transform of a Dirac delta function, it is given by

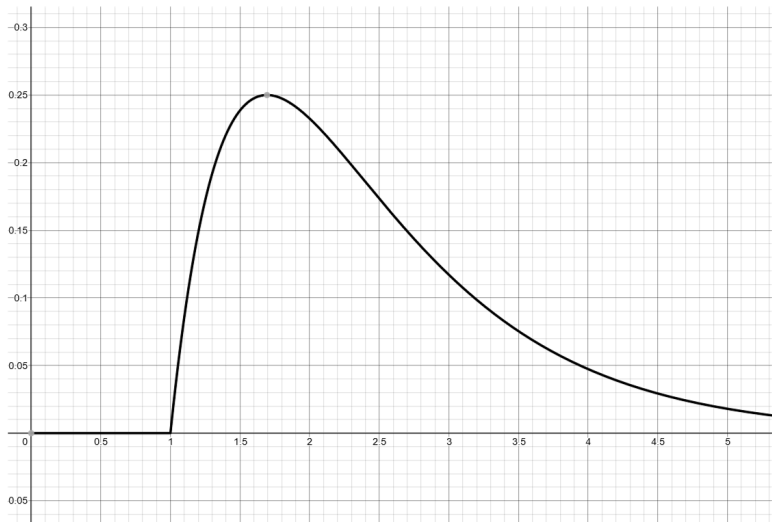
$$\mathcal{L}(\delta(t - a)) = e^{-as}.$$

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Example: Suppose we have a spring mass damper system with $m = 1$, $c = 3$ and $k = 2$.

The system is struck with a hammer at time $t = 1$. Find the response of the system.

Laplace Transforms



Exercises

- Find the Laplace transform of the function:

$$f(t) = \begin{cases} 0 & \text{if } t < 2 \\ t - 2 & \text{if } t \geq 2 \end{cases}$$

- Solve the initial value problem:

$$y'' + 3y' + 2y = 1, \quad y(0) = 0, \quad y'(0) = 0$$

using Laplace transforms.

Text Book References

Chapters 6.2, 6.3, 6.4